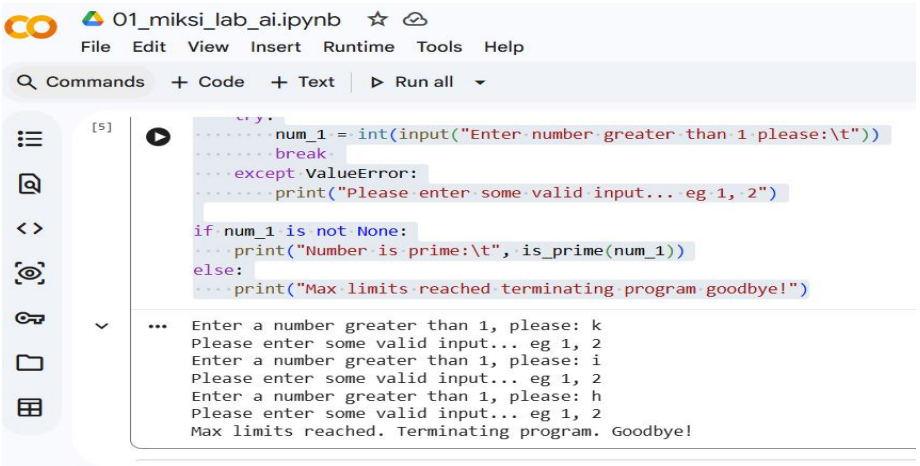
Name: Muhammad Ilyas
Reg No:L1F24BSAI0078
Section: AD1
AI LAB01
Task 01:
Source code:

```
# Write a program to check whether a number is prime.
# Note: Take input from user.

def is_prime(n):
    if n <= 1:
        print("Enter a number greater than 1.")
        return False

    for i in range(2, int(n**0.5) + 1):
        if n % i == 0:
            return False
    return True

limit = 0
num_1 = None

while limit < 3:
    limit += 1
    try:
        num_1 = int(input("Enter number greater than 1 please:\t"))
        break
    except ValueError:
        print("Please enter some valid input... eg 1, 2")

if num_1 is not None:
    print("Number is prime:\t", is_prime(num_1))
else:
        print("Max limits reached terminating program goodbye!")
```

Output:
Invalid case:



Valid case:

```
        print("Number is prime:", is_prime(num_1))
    else:
        print("Max limits reached terminating program goodbye!")
```

```
Enter number greater than 1 please:     k
Please enter some valid input... eg 1, 2
Enter number greater than 1 please:     3
Number is prime:         True
```

Task 2:

Source code:

```
#Take input from user and create a function to compute its factorial using a loop
## finding factorials
def finding_fctorial(n):
    if n < 0:
        return "number cant be negative less than zero(0)"
    sum = 1
    for i in range(2, n+1):
        sum = i * sum
    return sum

try:
    num_1 = int(input("enter number please..."))
    print("factorial is:", finding_fctorial(num_1))
except ValueError: # Fix 1: Corrected spelling
    print("invalid input>>>")
```

Output:

Invalid case:

```
try:
    num_1 = int(input("enter number please..."))
    print("factorial is:", finding_fctorial(num_1))
except ValueError: # Fix 1: Corrected spelling
    print("invalid input>>>")
```

```
enter number please...k
invalid input>>>
```

Valid case:

```
enter number please...5
factorial is: 120
```

Task 03:

Source code:

```
# Write a program to count vowels in a string.
def counting_vowels(strr):
    count = 0
    for ch in strr:
        if ch in 'aeiouAEIOU':
            count += 1
    return count
sentence = input("Enter  a sentence please...: ")
print("Vowels are :\t", counting_vowels(sentence))
```

Output:

```
Enter  a sentence please...: hi hello miksi how are you
Vowels are :     10
```

Alternative:

```
#alternative
# Write a program to count vowels in a string.
def counting_vowels(strr):
    count = 0
    for ch in strr:
        if (ch == 'a' or ch == 'A' or ch == 'e' or ch == 'E' or
            ch == 'i' or ch == 'I' or ch == 'o' or ch == 'O' or
            ch == 'u' or ch == 'U'):
            count += 1
    return count
sentance = input("Enter sentence: ")
print("Vowels are :\t", counting_vowels(sentance))
```

Output:

Same.

Task 04:

Source code:

```
# Create a list of 10 numbers and print even numbers only.
def show_datas(listt):
    print("showing complete list...")
    for i in range(len(listt)):
        print(f"on index {i} value is : \t{listt[i]} ")
def show_even(listt):
    print("showing only even from list...")
    for i in range(len(listt)):
        if(listt[i]%2==0):
            print(f"{listt[i]} on index number:\t {i} ")
my  list=[]
for i in range(10):
     num=int(input("enter number...eg 1,2..."))
     my  list.append(num)
show  datas(my  list)
show_even(my_list)
```

Output:

Entry:

```
enter number...eg 1,2...2
enter number...eg 1,2...2
enter number...eg 1,2...24
enter number...eg 1,2...44
enter number...eg 1,2...553
enter number...eg 1,2...34
enter number...eg 1,2...55
enter number...eg 1,2...33
enter number...eg 1,2...33
enter number...eg 1,2...33
```

Complete list show:

```
showing complete list...
on index 0 value is :   2
on index 1 value is :   2
on index 2 value is :   24
on index 3 value is :   44
on index 4 value is :   553
on index 5 value is :   34
on index 6 value is :   55
on index 7 value is :   33
on index 8 value is :   33
on index 9 value is :   33
```

Only even show:

```
showing only even from list...
2 on index number:       0
2 on index number:       1
24 on index number:      2
44 on index number:      3
34 on index number:      5
```

Same task with proper menu:

```python
# Create a list of 10 numbers and print even numbers only.
def show_datas(listt):
   print("showing complete list...")
   for i in range(len(listt)):
      print(f"on index {i} value is : \t{listt[i]} ")
def show_even(listt):
   print("showing only even from list...")
   for i in range(len(listt)):
      if(listt[i]%2==0):
         print(f"{listt[i]} on index number:\t {i} ")
my_list=[]
for i in range(10):
    num=int(input("enter number...eg 1,2..."))
    my_list.append(num)
choice=0
print("1 for show full list...")
print("2 for show even from list...")
print("3 for exit...")
choice=int(input("enter your choice please..."))
if(choice==1):
   show_datas(my_list)
elif(choice==2):
   show_even(my_list)
else:
  print("exiting....\ngoodbye have a good day dear user...")
```

Output:

```
1 for show full list...
2 for show even from list...
3 for exit...
enter your choice please...1
showing complete list...
on index 0 value is :   1
on index 1 value is :   2
on index 2 value is :   3
on index 3 value is :   4
on index 4 value is :   5
on index 5 value is :   6
on index 6 value is :   7
on index 7 value is :   8
on index 8 value is :   9
on index 9 value is :   10
```

Option 2:

```
1 for show full list...
2 for show even from list...
3 for exit...
enter your choice please...2
showing only even from list...
2 on index number:       1
4 on index number:       3
6 on index number:       5
8 on index number:       7
10 on index number:      9
```

Option 3:

```
1 for show full list...
2 for show even from list...
3 for exit...
enter your choice please...3
exiting....
goodbye have a good day dear user...
```

Task 05:

```
#Store student data in a dictionary and display formatted output
student_data = {
   "khuram": 85,
   "sofi": 92,
   "miksi": 78
}

def display  data(db):
   print("printing all students data...")
   for name,marks in db.items():
      print(f"Name is: {name} \t marks are: {marks}")

display_data(student_data)
```

Output:

```
display_data(student_data)
```

```
printing all students data...
Name is: khuram          marks are: 85
Name is: sofi    marks are: 92
Name is: miksi   marks are: 78
```

Task 06:

Source code:

```
#Remove duplicates from a list using sets.
orignal_data = [15, 20, 15, 30, 40, 20, 50]
set_data = list(set(orignal_data))
print(f"original list is:\t{orignal_data}")
print(f"set list is:\t{set_data}")
```

Output:

```
#Remove duplicates from a list using sets.
orignal_data = [15, 20, 15, 30, 40, 20, 50]
set_data = list(set(orignal_data))
print(f"original list is:\t{orignal_data}")
print(f"set list is:\t{set_data}")
```

```
original list is:       [15, 20, 15, 30, 40, 20, 50]
set list is:    [40, 15, 50, 20, 30]
```

Task 07:

Source code:

```
# Task 1: Create a tuple of 5 students
student_records = (
    (101, "khuram", (80, 75, 85)),
    (102, "sofi", (90, 92, 88)),
    (103, "miksi", (60, 65, 70)),
    (104, "artiza", (91, 92, 88)),
    (105, "baryal yousafzai", (76, 78, 80))
)


high_avg = 0.0
top_student = ""
above_75 = 0
print("printing studens averages...")
for student in student_records:
    name = student[1]
    marks = student[2]
    total_marks = marks[0] + marks[1] + marks[2]
    student_avg = total_marks / 3.0
    print(name, "average is:", student_avg)

    if student_avg > 75:
        above_75 = above_75 + 1
    if student_avg > high_avg:
        high_avg = student_avg
        top_student = name
```

```
print("Top student is\t:", top_student, "with score is:\t", high_avg)
print("Total above 75 student are\t:", above_75)


print("updating...")
temp_list = list(student_records)
temp_list[2] = (103, "miksi", (80, 85, 90))
updated_records = tuple(temp_list)

for data in updated_records:
    print(data)
```

Output:

```
printing studens averages...
khuram average is: 80.0
sofi average is: 90.0
miksi average is: 65.0
artiza average is: 90.33333333333333
baryal yousafzai average is: 78.0
Top student is  : artiza with score is:  90.33333333333333
Total above 75 student are      : 4
updating...
(101, 'khuram', (80, 75, 85))
(102, 'sofi', (90, 92, 88))
(103, 'miksi', (80, 85, 90))
(104, 'artiza', (91, 92, 88))
(105, 'baryal yousafzai', (76, 78, 80))
```

To solve repeating value issue:
Like 90.33333333333333
While printing we use {high_avg:2f} for formating

Task 08:
Short answers:
Difference between while loop and for loop:

A **while** loop runs until a condition becomes false. We often use it when we do not know how many times the loop will run while coding.

A for loop runs for a specific times of numbers in a proper sequence like a list or a string or a range.

Example:

while → run until condition changes

for → run fixed number of times

What is function modularity?

Function modularity means dividing a big/ huge complex program into small solveable functions.
Each function do one specific task of huge problem this approach makes the program easy to read easy to test and easy to reuse it.

Difference between list and tuple

A **list** is a collection of items that can be changed (mutable).
Example: we can add or remove or modify the elements.

A **tuple** is also a collection of items but cannot be changed after creation (immutable).

List uses square brackets [ ]

Tuple uses round brackets ( )

Why are sets used?

Sets are used to store unique values/elements only.
They automatically remove duplicate elements fro it.

Sets are very useful for fast searching engines and mathematical operations like union and intersection etc.

What is dictionary key restriction?

In a **dictionary** when working keys must be unique and immutable.

This means:

Two keys cannot be the same at atime.

Keys must be unchangeable types like int or strin, or tuple.

Mutable types like lists cannot be used as keys.